

[VJC/2025/Q2]

2 Name your Jupyter Notebook as:

TASK2_<your name>_<class>_<index number>.ipynb

A program is required to manage an ordered linked list of student records, where each record contains a student's ID (integer) and name (string). The linked list must be maintained in ascending order based on the student's ID. To optimise memory usage, a free space list will be implemented to reuse deleted nodes.

The class Node contains three attributes:

- `student_id` the integer ID of the student, initialised to -1
- `student_name` the string name of the student, initialised to empty string
- `next_pointer` points to the index of the next node in the array, initialised to -1.

The class Node has the following methods:

- `set_id()` that assigns a value to `student_id`
- `set_name()` that assigns a value to `student_name`
- `get_id()` that returns `student_id`
- `get_name()` that returns `student_name`

The class LinkedList contains the three attributes:

- `nodes_array` a list of Node objects, with size 10
- `start_pointer` points to the index of the first node in the linked list, initialised to -1
- `next_free_pointer` points to index of the first available node in the free space list, initialised to 0.

The class LinkedList has the following methods:

- a constructor that
 - initialise `nodes_array` as a list of 10 Node objects.
 - link all nodes into the free space list by setting `next_pointer` of each node to the next index (e.g., node 0 points to 1, node 1 points to 2, etc.), with the last node's `next_pointer` set to -1.
 - initialise `start_pointer` to -1 and `next_free_pointer` to 0.
- `insert_record(student_id, student_name)` to insert a record into the ordered linked list.
- `delete_record(student_id)` to delete a record from the linked list and return the node to the free space list.
- `display_records()` to output all records in the linked list in order. The output should look like the following:
Student ID: <insert student_id>, Name: <insert student_name>

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 2.1  
Program code
```

Output:

[VJC/2025/Q2 Cont ']

Task 2.1

Write program code to declare the class `Node`, its constructor and its methods. [4]

Task 2.2

Write program code to declare the class `LinkedList` and its constructor.. [4]

Task 2.3

Write the `display_records()` method for the `LinkedList` class. The method should output all records in the linked list in ascending order of `student_id`. The output for each record should be in the following format:

```
Student ID: <insert student_id>, Name: <insert student_name>
Student ID: <insert student_id>, Name: <insert student_name>
```

 [2]

Task 2.4

Write the `insert_record(student_id, student_name)` method for the `LinkedList` class. The method should:

- Check if the free space list has a next available node
- If there is a next available node in the free space list,
 - store the new record in this node, remove it from the free space list and insert it into the ordered linked list at the correct position
 - update `next_free_pointer` and `start_pointer` to appropriate values
- If there is no available node in the free space list, output an appropriate message. [6]

Test your program by inserting the following records in order and then, calling `display_records()`:

- (101, "Alice")
- (103, "Bob")
- (102, "Charlie") [3]

Task 2.5

Write the `delete_record(student_id)` method for the `LinkedList` class. The method should:

- Search for the record with the given `student_id`
- Remove the node from the linked list and insert it to the start of the free space list
- Update `next_free_pointer` and `start_pointer` to appropriate values
- Output "Record deleted" if successful, or "Record not found" otherwise. [6]

Test your program by:

- Deleting the record with `student_id = 102`
- Deleting the record with `student_id = 104`

Show the output of `display_records()`. [1]

Save your Jupyter Notebook for Task 2.